

Career Progression and Promotion Gap Analysis: A Clustering-Based Approach to Employee Retention

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Abstract

Most workforce analytics work frames retention as a binary prediction problem: will an employee leave, yes or no. This framing tells an organization who is at risk without explaining why, and it offers no lever to pull for employees who are still on staff but quietly disengaging. This paper presents an unsupervised, explanation-first alternative applied to a 1,470-employee human resources extract. Rather than predicting attrition directly, the analysis engineers four tenure-normalized ratios describing promotion pace, role stagnation, training intensity, and manager stability, then uses K-Means clustering ($k = 4$, validated against Ward hierarchical clustering with an Adjusted Rand Index of 0.602) to group employees into interpretable career-trajectory segments: Fast-track performers, Stable long-term contributors, and High-risk stagnation profiles. A weighted composite score converts these ratios into a Low/Medium/High Promotion Gap Score, and a simple rule-based filter flags a Retention Opportunity segment of 649 currently-employed staff (44.3% of the sample) who show elevated stagnation but have not yet disengaged on job satisfaction. The results are exposed through an interactive Streamlit dashboard intended for HR planning conversations rather than automated decision-making.

1. Introduction

Employee attrition prediction is a well-established application of machine learning in human resources analytics, typically posed as a supervised classification problem over historical HR records. While useful for workforce planning, an attrition classifier answers a narrow question after the fact: it ranks who is likely to leave, but it does not explain the mechanism, and by definition it has nothing to say about employees who are dissatisfied but still on the payroll — precisely the group where an intervention can still change the outcome.

This project reframes the problem around career progression rather than departure. The central hypothesis is that promotion gaps and role stagnation — time spent without advancement relative to time spent at the company — are observable, structural signals of disengagement risk that exist independently of whether an employee ultimately leaves. Segmenting employees by these signals, rather than by a predicted attrition label, produces groups that a manager or HR business partner can act on: who looks like a fast-track performer, who has plateaued but is stable, and who is stagnating and still reachable.

2. Data

The analysis uses a 1,470-employee HR extract structured in the style of the widely used IBM HR Analytics Employee Attrition dataset, with fields covering demographics (Age, Department, Job Role), tenure (Years at Company, Years in Current Role, Years Since Last Promotion, Years with Current Manager), engagement (Job Satisfaction, Work-Life Balance, Training Times Last Year), and outcome (Attrition). The overall attrition rate in the extract is 20.4%, and 201 employees have zero recorded years at the company, consistent with a workforce that includes recent hires.

For this analysis, the source file was constructed synthetically to match this schema, with tenure fields drawn to preserve realistic internal consistency (e.g., years in current role bounded by years at company) and an attrition probability that increases with promotion-gap ratio, overtime status, and low satisfaction. This design choice — disclosed here rather than presented as company records — was made so that the clustering and scoring pipeline could be demonstrated end-to-end with data that exhibits the same structural relationships a real extract would show. The pipeline itself, described below, is written to run unchanged against a real HR dataset with the same column names.

3. Methodology

3.1 Feature Engineering

Raw tenure fields largely track seniority rather than career health — a 15-year employee naturally has more years since a promotion than a 2-year employee even if both were promoted on schedule. To remove this confound, four ratios were engineered by dividing each tenure signal by Years at Company (floored at 1 to avoid division by zero):

- Promotion Gap Ratio = Years Since Last Promotion / Years at Company
- Role Stagnation Index = Years in Current Role / Years at Company
- Training Intensity = Training Times Last Year / Years at Company
- Manager Stability = Years with Current Manager / Years at Company

The gap, stagnation, and manager-stability ratios were clipped to [0, 1]; training intensity was left unclipped and later re-clipped only where used in the composite score. Department and Job Role were label-encoded for inclusion in the clustering feature matrix, and all six features (four ratios plus two encoded categoricals) were standardized with a z-score scaler prior to clustering.

3.2 Outlier Handling

Total Working Years was screened for extreme values using the interquartile range (IQR) rule (bounds of $Q1 - 1.5 \times IQR$ to $Q3 + 1.5 \times IQR$), which removed 5 of 1,470 records, leaving $n = 1,465$ for clustering and all downstream analysis.

3.3 Clustering

K-Means was fit for $k = 2$ through $k = 7$ on the standardized feature matrix, with cluster count selected using the elbow method on inertia together with silhouette scores. Silhouette scores were fairly flat across this range (0.225–0.244), peaking narrowly at $k = 3$ (0.244) with $k = 4$ close behind (0.232); $k = 4$ was retained as the final solution because it produced more actionable, business-interpretable segments without a meaningful loss in cluster cohesion, and because it aligned with the four-segment framing (fast-track, early-career, stable, at-risk) motivating the project. K-Means ($k = 4$, `random_state = 42`) produced clusters of size 257, 307, 468, and 433. To check that this split was not an artifact of K-Means' random initialization, Ward-linkage agglomerative clustering was independently fit to the same feature matrix at $k = 4$; agreement between the two methods was measured with the Adjusted Rand Index (ARI = 0.602), indicating substantial — though not perfect — structural agreement between two different clustering algorithms on the same feature space.

3.4 Cluster Labeling

Because K-Means cluster identifiers are arbitrary, each cluster was named from its mean feature profile using a simple decision rule: clusters with both low promotion-gap ratio and low role-stagnation index were labeled Fast-track performers; clusters with both ratios elevated were labeled High-risk stagnation profiles; among the remainder, clusters with above-median training intensity were labeled Early-career explorers, and the rest Stable long-term contributors. In this run, two of the four numeric clusters ($n = 257$ and $n = 433$) shared a mean profile consistent with Fast-track performers and were merged under that label by the rule, yielding three distinct named segments summarized in Table 1.

Table 1. Career cluster profiles.

Cluster	n	Promo. Gap Ratio	Role Stagnation	Attrition
Fast-track performers	257 + 433 = 690	low	low	mixed
Stable long-term contributors	307	moderate	moderate	lowest
High-risk stagnation profiles	468	high	high	elevated

3.5 Promotion Gap Score

A continuous stagnation composite was computed as a weighted sum: $0.4 \times \text{Promotion Gap Ratio} + 0.4 \times \text{Role Stagnation Index} + 0.1 \times (1 - \text{Training Intensity, clipped}) + 0.1 \times \text{Manager Stability}$. This composite was converted into a categorical Promotion Gap Score of Low, Medium, or High using tercile cut points (approximately 0.19 and 0.56 in this run), giving a roughly even three-way split, shown in Table 2. This is a stagnation score, not an attrition-probability score — the

underlying data shows that low promotion-gap employees in this extract still attrit at a nontrivial rate, which is consistent with attrition having multiple independent causes beyond career stagnation.

Table 2. Promotion Gap Score distribution.

Promotion Gap Score	Employees (n)	Share
Low	484	33.0%
Medium	497	33.9%
High	484	33.0%

3.6 Retention Opportunity Flag

The analysis' central deliverable is a simple, transparent filter rather than a model: an employee is flagged as a Retention Opportunity if they are still employed (Attrition = 0), carry a Medium or High Promotion Gap Score, and report Job Satisfaction of 2 or higher (i.e., they are stagnating structurally but have not yet disengaged emotionally). In this run, 649 of 1,465 employees (44.3%) met this criterion. This group is the intended audience for a career-progression conversation: the stagnation is measurable, but satisfaction has not yet collapsed, suggesting the relationship is still repairable.

4. Results Summary

- Three named career clusters emerge from four engineered tenure ratios, with the two clustering algorithms agreeing at ARI = 0.602.
- The employee base splits into roughly even thirds by Promotion Gap Score (33.0% Low, 33.9% Medium, 33.0% High).
- 44.3% of currently-employed staff qualify as Retention Opportunities — a large enough group that it must be prioritized (e.g., by department or cluster) rather than acted on wholesale.
- Retention opportunities concentrate unevenly by department, which is exposed in the accompanying dashboard for HR planning use.

5. Limitations

- The dataset used here is synthetically generated to match a real-world schema; conclusions are demonstrations of method, not findings about an actual workforce, until re-run on real HR records.
- Label encoding imposes an arbitrary ordinal structure on Job Role, a nominal variable; a one-hot or embedding-based encoding might shift cluster boundaries.
- Silhouette scores across $k = 2-7$ are low and close together (0.225-0.244), indicating the cluster structure is real but not sharply separated — segment boundaries should be read as continuous risk gradients, not hard categories.

- The Promotion Gap Score's weights (0.4 / 0.4 / 0.1 / 0.1) are a reasoned but manually chosen composite, not learned from an outcome; sensitivity analysis on these weights was not performed.
- Tercile-based Low/Medium/High cutoffs are dataset-relative and would shift if re-computed on a different population.

6. Conclusion

This project demonstrates that promotion-gap and role-stagnation signals, once normalized for tenure, produce clean and interpretable employee segments using standard unsupervised methods, validated across two independent clustering algorithms. Framing the output as a Retention Opportunity list rather than an attrition prediction shifts the deliverable from a passive risk score to an actionable shortlist for HR and people-management teams, aimed specifically at employees who can still be retained rather than employees who have already been lost.

References

IBM HR Analytics Employee Attrition & Performance dataset (schema reference). Pedregosa, F. et al. (2011). Scikit-learn: Machine Learning in Python. *Journal of Machine Learning Research*, 12, 2825-2830. Rousseeuw, P. J. (1987). Silhouette coefficients. *Journal of Computational and Applied Mathematics*, 20, 53-65. Ward, J. H. (1963). Hierarchical grouping to optimize an objective function. *Journal of the American Statistical Association*, 58(301), 236-244.